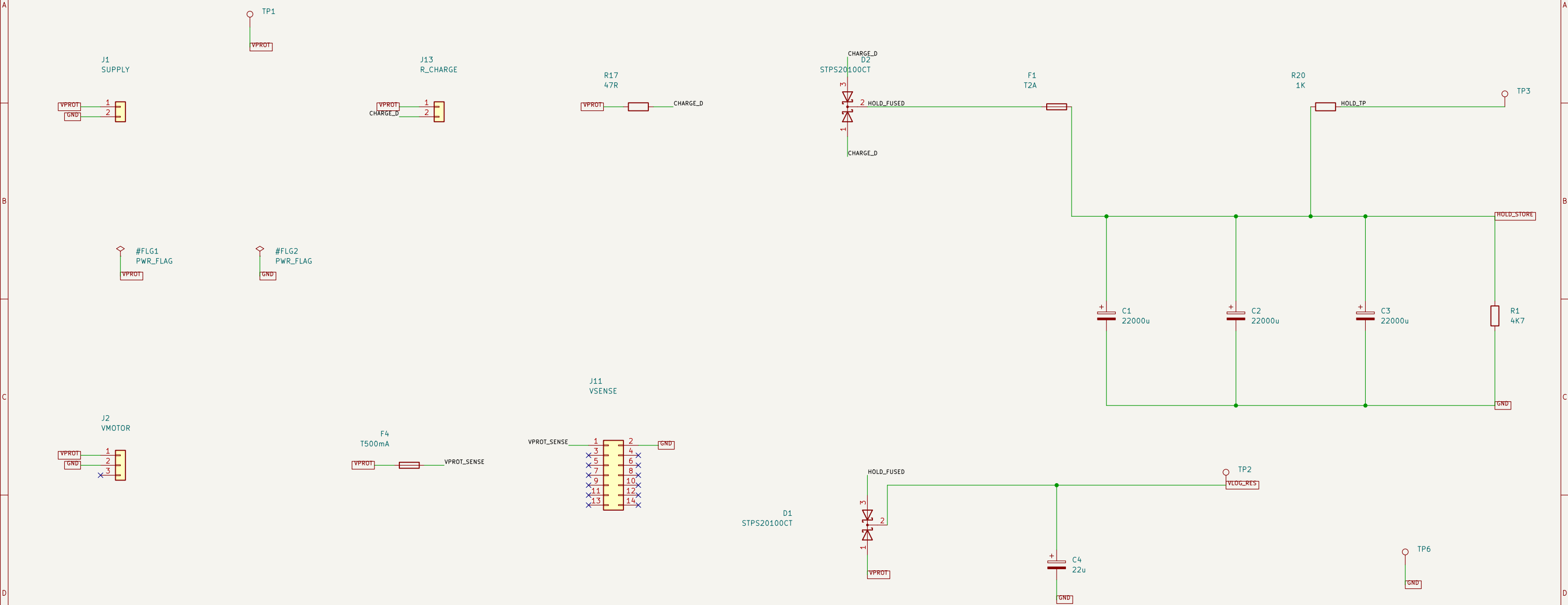
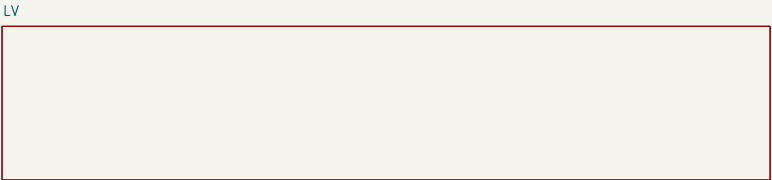




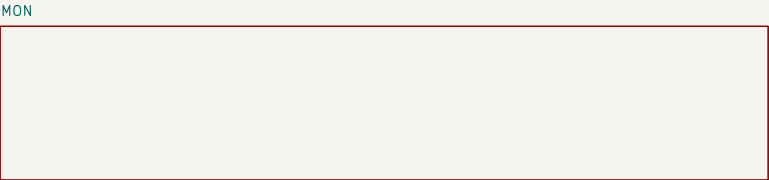
R17 poza PCB: HSA2547RJ na osobnej blasze. J13 jest zakonczeniem jego przewodow.

TP3 przez R20=1k/2W. DMM >=10 Mohm. Bank 3x22mF: do 41J przy 32V, rozladowanie przez R1 do 22 min.

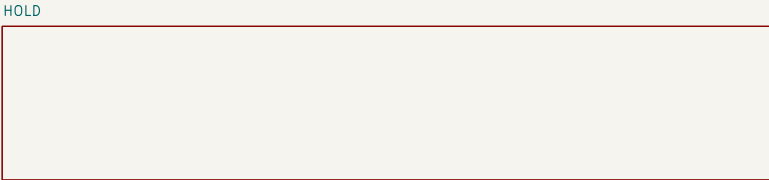
F1 T2A, F4 T0,5A: Schurter SPT 5x20, 300 VDC. VSENSE niesie ok. 25 uA; F4 chroni tylko przewod.



Plik: LV.kicad_sch



Plik: MON.kicad_sch



Plik: HOLD.kicad_sch

VPROT <= 5 A system budget; HOLD 50 ms at <= 6 W; review before Gerbers

Prototype / schematic + PCB review

Sheet: /
File: P02.kicad_sch

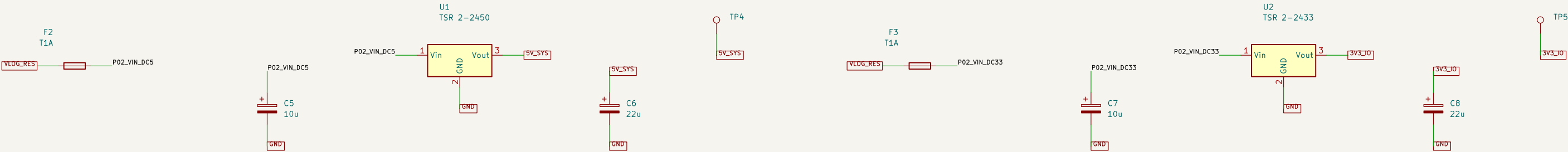
Title: P02-R3 / P02

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-26

Rev: P02-R3

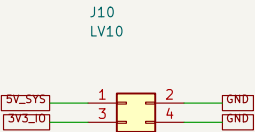
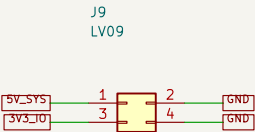
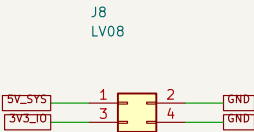
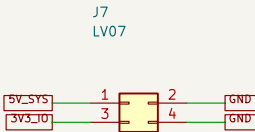
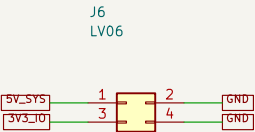
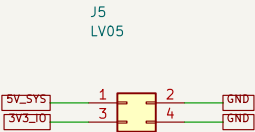
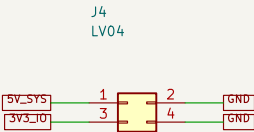
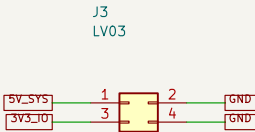
Id: 1/4



BUDZET LACZNY <=6W na VLOG_RES (wliczaj straty przetwornic). Nie 2A na kazde zlacze.

LV03..LV10: 1=5V_SYS, 2=GND, 3=3V3_I0, 4=GND. Nie laczyc 3V3_I0 z 3V3_CORE.

F2/F3: Schurter 0001.2504 T1A, 300 VDC – zwloczne, jak zaleca TRACO dla TSR2.



Dodatkowa pojemnosc obciazenia: suma <=600uF na 5V i <=900uF na 3V3 (wlicz C6/C8).

VPROT <= 5 A system budget; HOLD 50 ms at <= 6 W; review before Gerbers

Prototype / schematic + PCB review

Sheet: /LV/

File: LV.kicad_sch

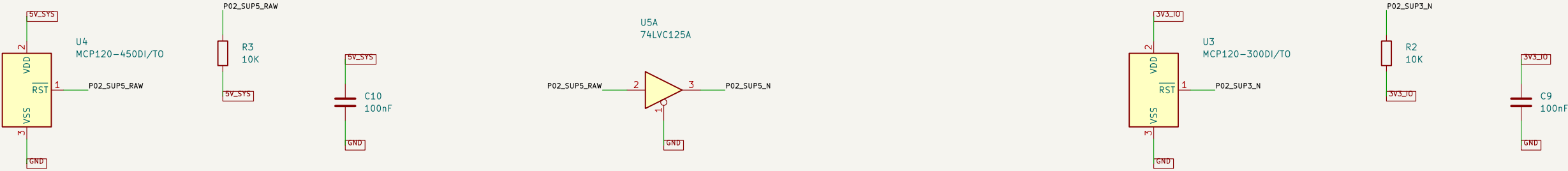
Title: P02-R3 / LV

Size: A3 Date: 2026-09-26

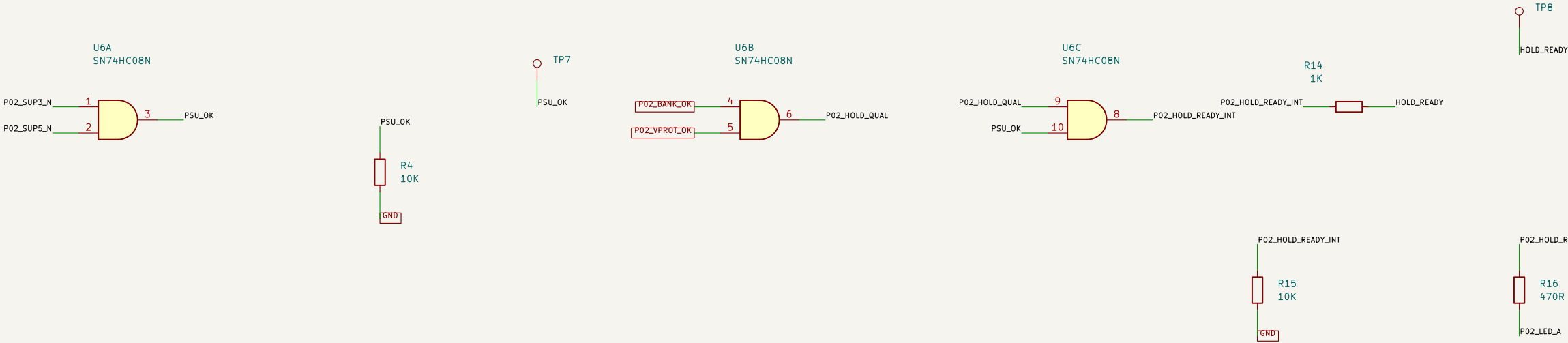
Rev: P02-R3

KiCad E.D.A. 10.0.6

Id: 2/4

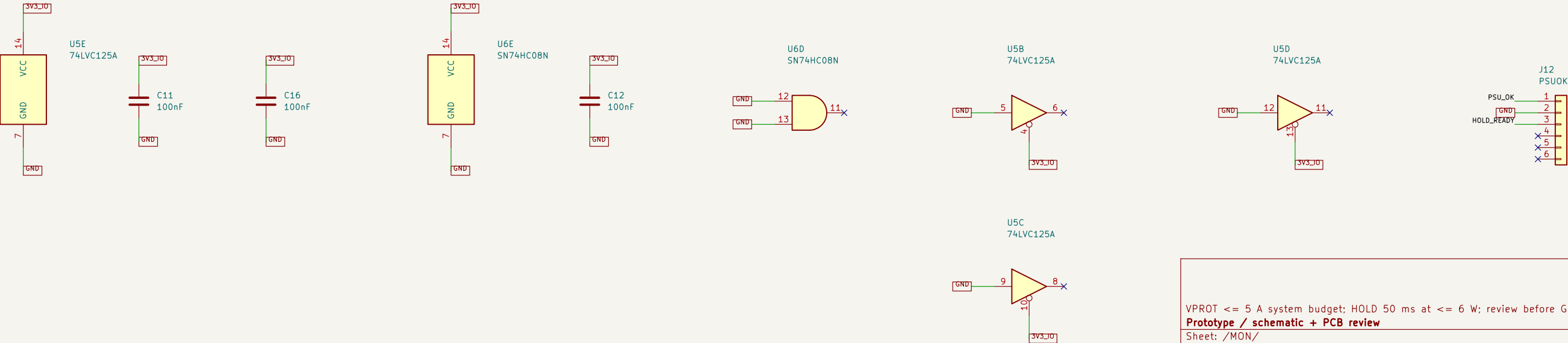


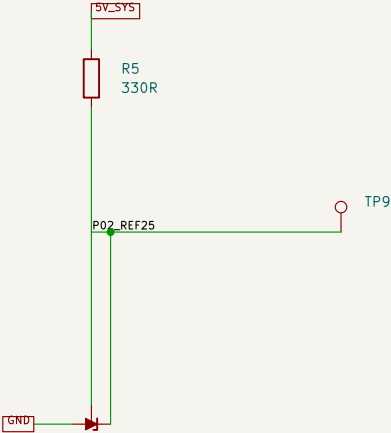
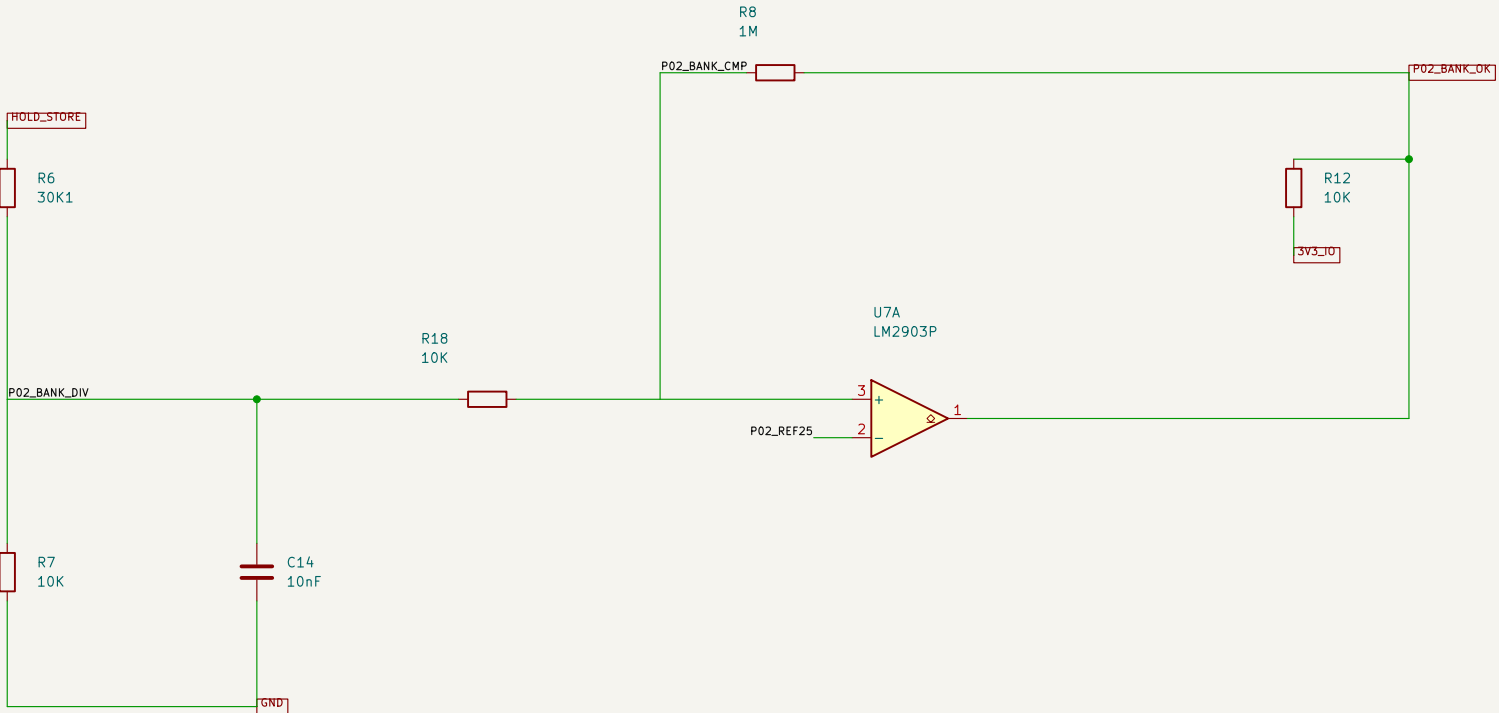
U5A: wejście toleruje 5V; zasilanie 3V3. C16 na adapterze przy samym IC (nie na płycie P02).



HOLD_READY = BANK_OK & VPROT_OK & PSU_OK. Sygnał lokalny; J12.3 na P04 nadal NC.

15 s kwalifikacji wykonuje operator. CORE nie realizuje tej funkcji w tej rewizji.

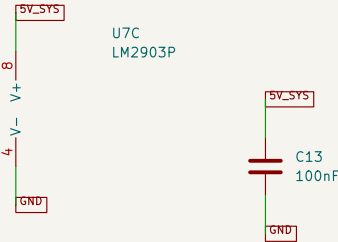
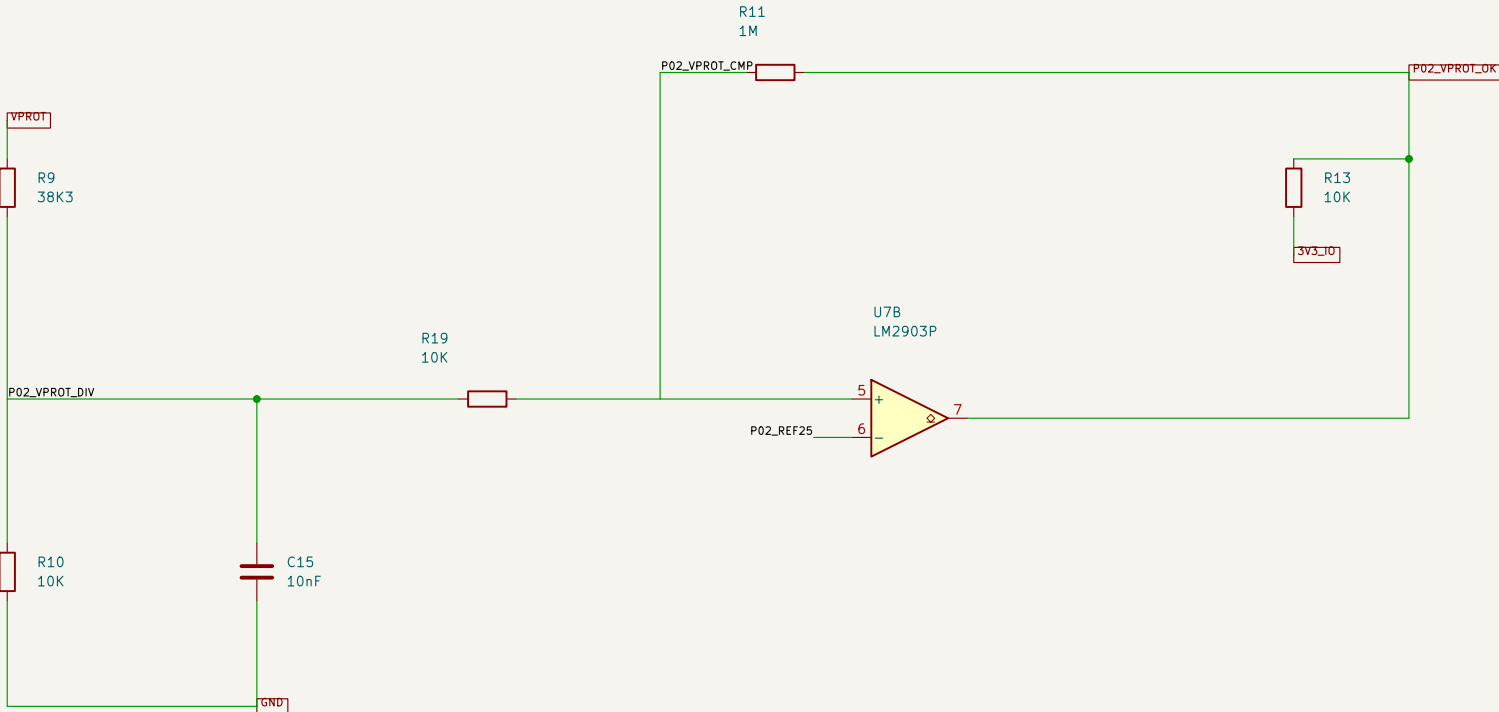




U8 TI LP: 1=K, 2=A, 3=REF.

R6/R7/R9/R10: 0.1%, 25ppm/K.

R5: 330R, IKA ok. 7.6mA.



FILTR: C14/C15 PRZED R18/R19. Sprzezenie R8/R11 za nimi, przy wejsciu komparatora.

Progi z tolerancjami: docs/HOLD-ANALIZA.md. Przerwany F1: LED gasnie po ok. 1-2 min (R1 rozladowuje bank). CeFF nie jest sprawdzane.